SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct: (Y6S200103)

I Condace Carol *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ~~Cattu~~ .

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	2	2
Application & Analysis	20%	1	1	2
Explanation & Clarity	15%	—	—	—

Total Marks 22 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

vivo V20 SE

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:
DataLink→MACAddress→LocalDelivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:
ErrorControl
→DataLinkLayer→ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:
Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗
Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

vivo V20 SE

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

vivo V20 SE

LO1 - AT-2

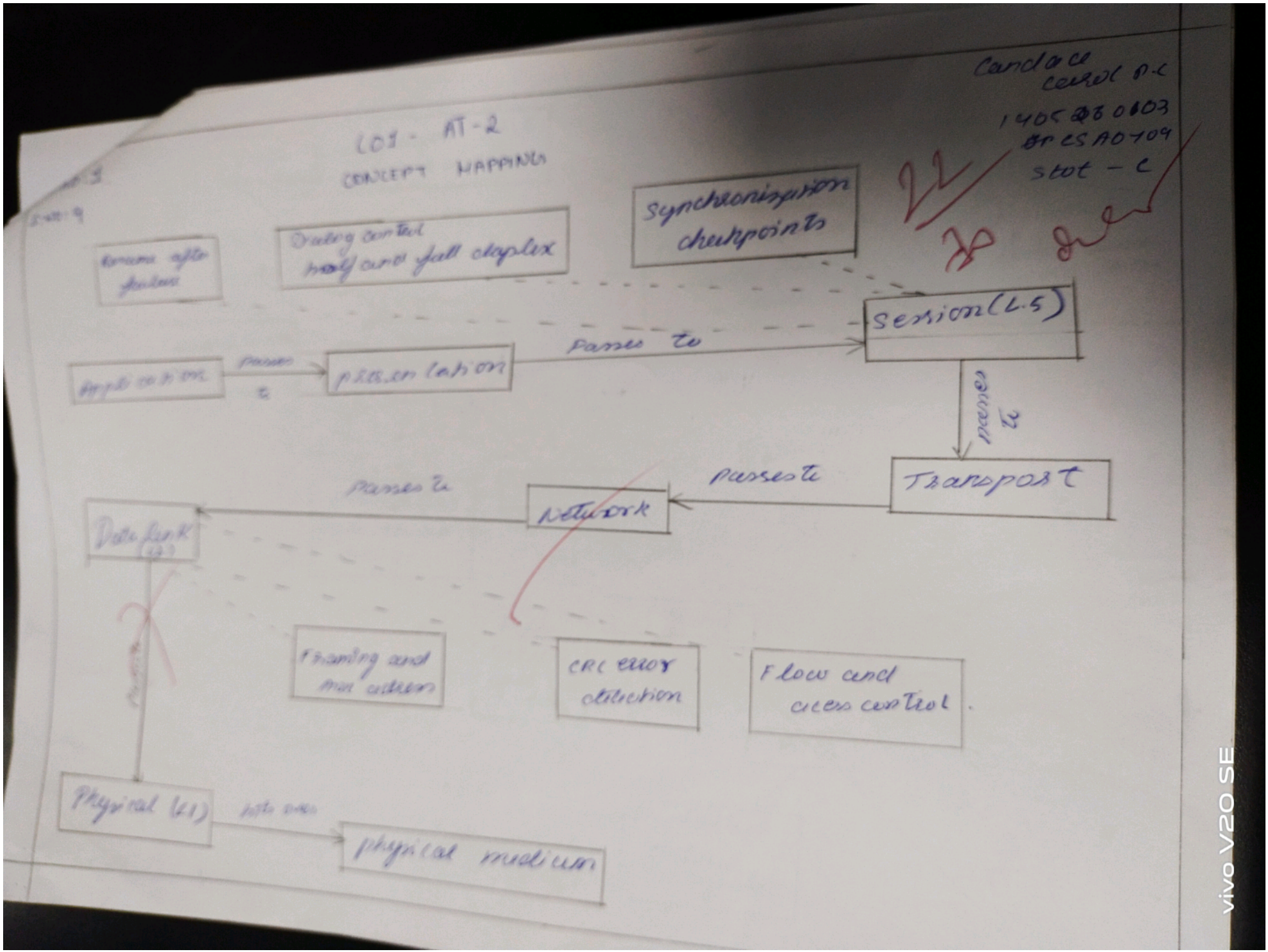
CONCEPT MAPPING

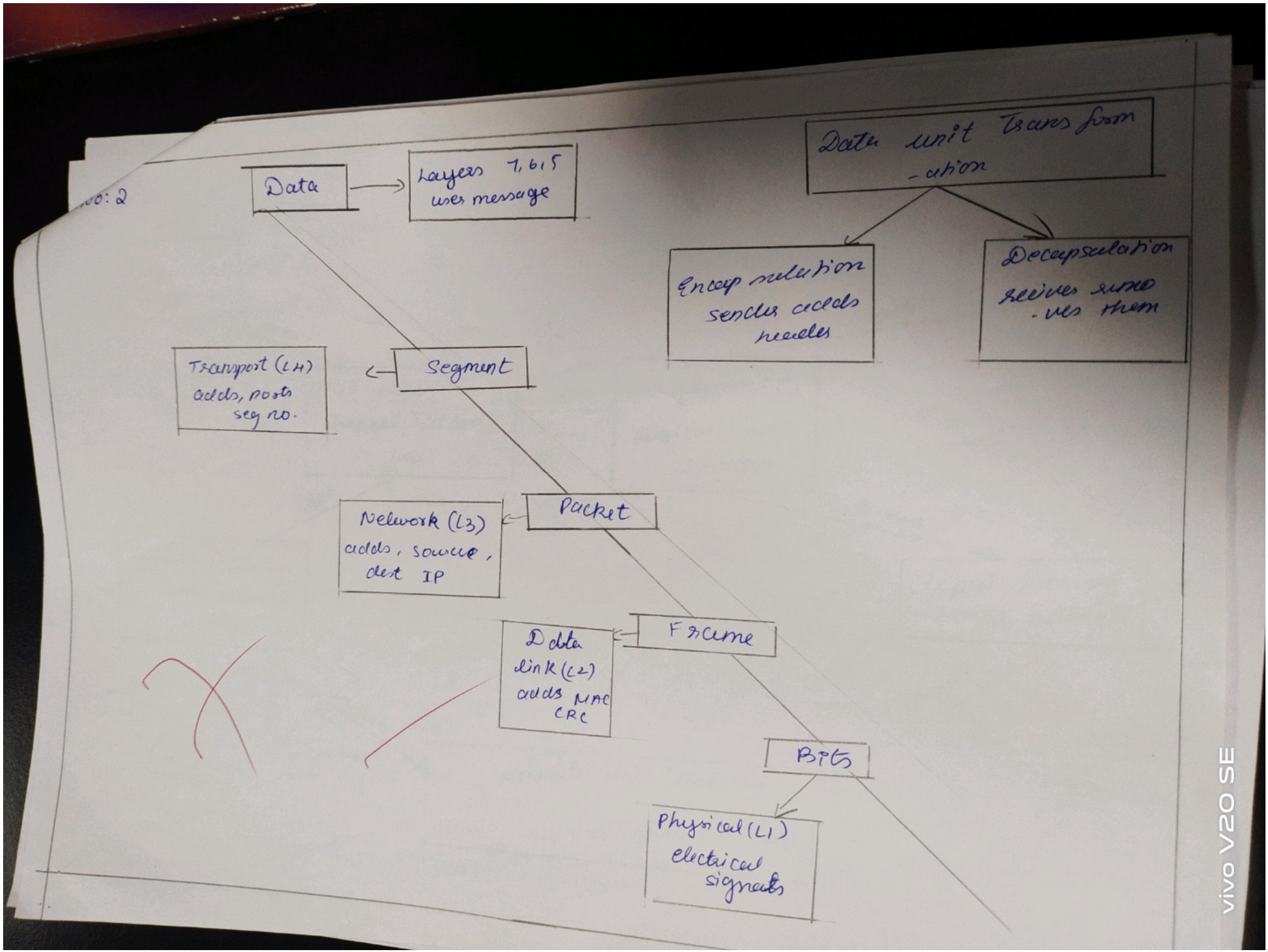
Candace Carol D.C

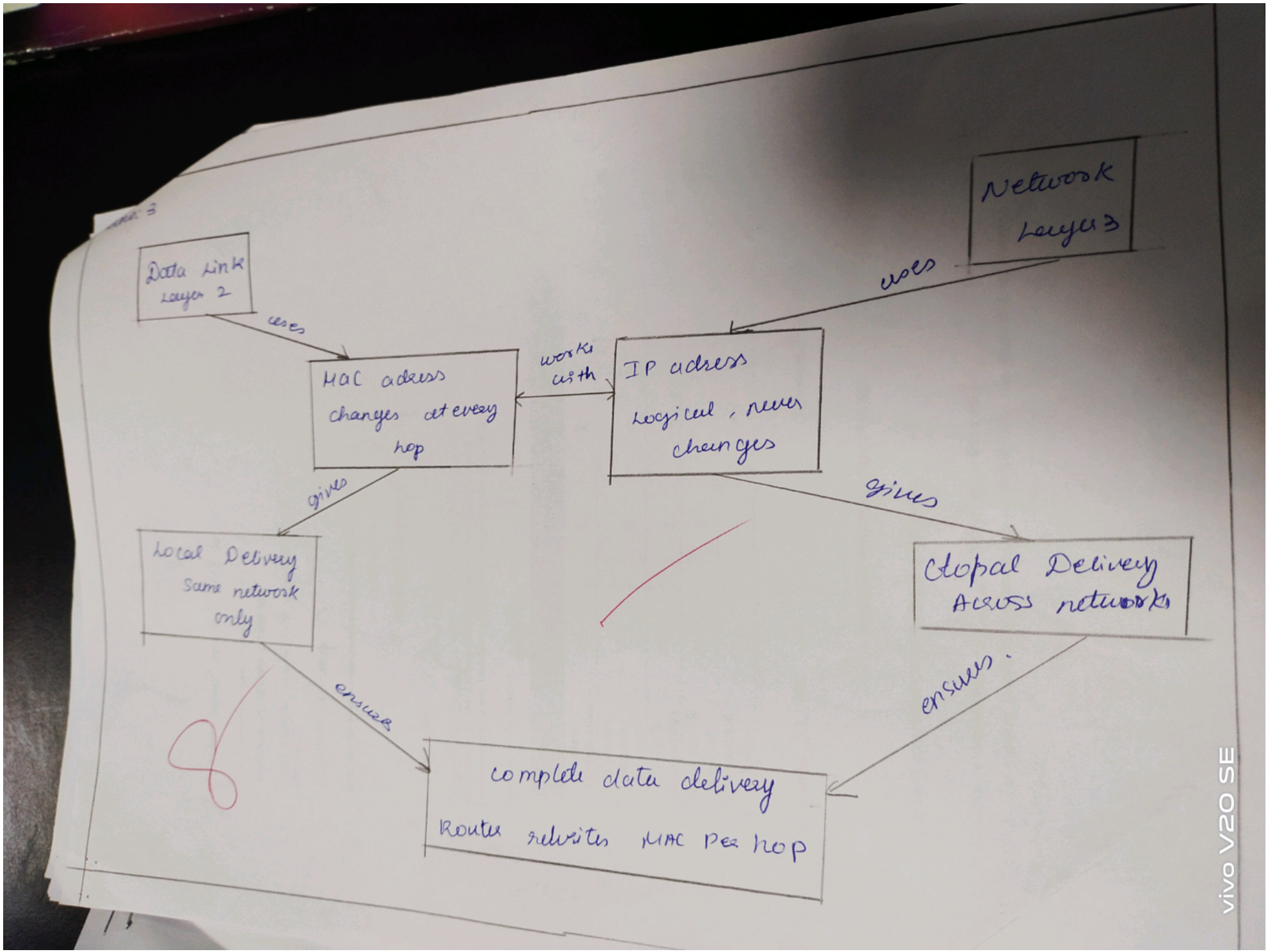
140C@80103

@rCSA0709

Slot - C




Data unit Transformation
Data
Layers 7,6,5 uses message
Encapsulation sends adds headers
Decapsulation receives removes them
Segment
Transport (L4) adds, ports seq no.
Packet
Network (L3) adds, source, dest IP
Frame
Data link (L2) adds MAC CRC
Bits
Physical (L1) electrical signals


Network Layer 3
Data Link Layer 2
uses
uses
MAC address changes at every hop
works with
IP address Logical, never changes
gives
gives
Local Delivery Same network only
Global Delivery Across networks
ensures
ensures
complete data delivery
Router rewrites MAC per hop